

Chapter 136 — Benign Renal Tumours

- **Incidence has risen with incidental detection:** wider use of abdominal imaging has increased the number of benign renal tumours found.
- **Up to about a quarter of surgically treated small renal masses (SRMs) prove benign** on final pathology. This challenges routine surgery for every suspicious lesion, given the morbidity of intervention.
- **The benign differential of the SRM:** renal cysts of varying anatomical complexity; papillary adenoma; oncocytoma; metanephric tumours (metanephric adenoma, metanephric adenofibroma, metanephric stromal tumour); mixed epithelial and stromal renal tumours; and adult renal mesenchymal tumours — angiomyolipoma (AML), renal haemangioblastoma, juxtaglomerular cell tumour and renomedullary interstitial cell tumour.
- **No imaging modality fully separates benign from malignant** — not ultrasonography (US), computed tomography (CT) or even magnetic resonance imaging (MRI).
- **Two benign lesions are usually diagnosable on imaging:** the simple cyst (smooth, non-enhancing wall) and AML (macroscopic fat). **Clinical pointers to benignity** — small size, female sex, older age — are not reliable, so resection is sometimes required for diagnosis.
- **Renal mass biopsy** spares intervention in a significant proportion of patients with benign histology; the American Urological Association (AUA) guideline gives it a role in SRM decision-making.
- **Management follows the perceived risk of renal cell carcinoma (RCC) and metastasis:** no dedicated follow-up (simple cyst) → selective embolisation (larger AML) → surgical extirpation (solid or complex cystic mass).

RENAL CYSTS

Epidemiology, Pathophysiology and Natural History

- **Fluid-filled masses, usually asymptomatic** and found incidentally as SRMs.
- **Common, commoner in men, and increasing with age:** about one in ten trauma CT scans shows an incidental cyst; on population MRI about one-third of men and one-fifth of women have cysts, rising to around half of men and over two-fifths of women aged ≥70 years. Average size is under 1.5 cm.
- **Risk factors:** increasing age, male sex and renal dysfunction.
- Larger cysts only infrequently cause flank or abdominal pain, a palpable mass or haematuria.
- **Mostly sporadic with a benign natural history;** slow growth of about **1–2 mm per year**.

Evaluation

- Most incidental masses on non-renal-protocol imaging are determinate; **cysts form the majority of benign renal masses**.
- **Simple cyst:** any size, homogeneous low attenuation ≤20 Hounsfield units (HU) (fluid density) → no further imaging.
- **Non-simple features raise concern for malignancy** — heterogeneity, attenuation >20 HU, septa, wall thickening, mural nodules, calcification — and prompt further imaging and Bosniak stratification.
- **Bosniak classification:** described by Morton Bosniak in 1986 on CT criteria; later refined to add **category IIF** (“F” = follow-up); updated as **version 2019** to include MRI, chiefly to address the high proportion of benign findings among resected Bosniak III lesions.

- **Performance** (systematic reviews): **high sensitivity, modest specificity** for malignancy. Malignancy is uncommon in I, II and IIF, about half of III and the great majority of IV.
- **Bosniak III has a low positive predictive value (PPV)** → substantial overtreatment. If the outcome that matters is avoidance of metastasis, the PPV is very low and the number needed to treat very high — supporting **surveillance of Bosniak III**.

TABLE 136.1 THE UPDATED BOSNIAK CLASSIFICATION SYSTEM FOR CYSTIC RENAL MASSES

Category	Defining imaging features (version 2019)	Risk of malignancy
I	Well-defined, thin (≤ 2 mm) smooth wall; homogeneous simple fluid; no septa; no calcification; wall may enhance	Benign
II	All well-defined with thin (≤ 2 mm) smooth walls. Three types: (a) few (≤ 3) thin (≤ 2 mm) enhancing septa, any non-enhancing septa, calcification of any type; (b) homogeneous, markedly T2-hyperintense on non-enhanced MRI; (c) homogeneous, markedly T1-hyperintense on non-enhanced MRI	Highly likely benign
IIF	Two types: (a) smooth, minimally thickened (3 mm) enhancing wall, or smooth minimal thickening (3 mm) of ≥ 1 enhancing septum, or many (≥ 4) smooth thin enhancing septa; (b) heterogeneously hyperintense on unenhanced fat-saturated T1-weighted MRI	Likely benign, nearly always indolent; serial follow-up to exclude malignancy
III	≥ 1 enhancing thick (≥ 4 mm) or enhancing irregular (≤ 3 mm obtuse-margined convex protrusions) wall or septum	Significant (about 50%)
IV	≥ 1 enhancing nodule : ≥ 4 mm convex protrusion with obtuse margins, or a convex protrusion of any size with acute margins	Likely malignant (about 90%)

Data from Silverman et al., 2019.

- **What it is for**: sorts cysts into those needing nothing (I–II), surveillance (IIF) and a management decision (III–IV).
- **The pattern**: wall and septal thickness escalate ≤ 2 mm → 3 mm → ≥ 4 mm; then irregularity; then a nodule.
- **Irregularity versus nodule**: an obtuse-margined protrusion ≤ 3 mm is irregularity (III); ≥ 4 mm obtuse-margined, or any acute-margined protrusion, is a nodule (IV).
- **Traps**: calcification of any type is compatible with Bosniak II; a thin wall may enhance in Bosniak I; few (≤ 3) thin septa is II, many (≥ 4) is IIF.
- **Figures 136.1–136.2** (CT): II with internal calcification; IIF with several thin septations; III with thick irregular septations; IV with a solid enhancing nodule — the bivalved specimen proved conventional RCC.

Management

- **Canadian Urological Association (CUA) 2023 guideline** endorses Bosniak version 2019; developed with a modified Grading of Recommendations, Assessment, Development and Evaluation (GRADE) approach.
- **Bosniak I and II: growth is expected** and is not a sign of malignancy; malignant transformation is exceedingly rare.
- **Large (≥ 3 cm) hyperdense cysts**, formerly Bosniak IIF, are no longer IIF under version 2019.
- **Shared decision-making** governs the III–IV choice, weighing circumstances, values and preferences.
- **Cystic RCC has a more favourable natural history than solid RCC**: much of the measured volume is fluid, so tumour burden is lower and size correlates poorly with outcome → proposed stage **pT1c** (“c” = **cystic**).
- **Surveillance of cystic masses up to 4 cm** parallels AUA surveillance of solid SRMs presumed to be RCC; evidence is from small retrospective series, with prospective studies ongoing.

GUIDELINE RECOMMENDATIONS FOR CYSTIC RENAL MASSES

Category / size	Recommendation	Body
Bosniak I and II	No follow-up	CUA 2023
Bosniak IIF	Active surveillance for up to 5 years	CUA 2023
Bosniak III–IV ≤2 cm	Active surveillance preferred	CUA 2023
Bosniak III–IV 2–4 cm	Surveillance or surgical excision (shared decision)	CUA 2023
Bosniak III–IV >4 cm	Surgical removal ; partial preferred over radical nephrectomy when technically feasible	CUA 2023
Bosniak IV	Manage as a localised solid renal mass	EAU 2023
Bosniak IV, significant solid component	Biopsy may have a role if the result could alter management (expert opinion)	CUA 2023
Thermal ablation	Effectiveness for cystic masses uncertain ; limited studies	—

- EAU = European Association of Urology. **Pattern:** size, not category alone, drives the III–IV decision; the ≤4 cm surveillance window mirrors solid SRM practice.

Autosomal Dominant Polycystic Kidney Disease

- **Autosomal dominant polycystic kidney disease (ADPKD):** systemic disorder with progressive cystic transformation of both kidneys → gross enlargement and distortion; **hypertension from raised plasma renin and aldosterone and intrarenal ischaemia.**
- **Slowly progressive chronic kidney disease (CKD); one of the commonest causes of end-stage renal disease (ESRD);** about 12.5 million people worldwide, about 1 in 1000–2500.
- **Extrarenal cysts:** liver, pancreas, spleen, ovaries, seminal vesicles, arachnoid.
- **Genetics: PKD1 (chromosome 16) → polycystin-1; PKD2 (chromosome 4) → polycystin-2.** These integral membrane proteins regulate ciliary function on tubular cells; the two are proposed to form a receptor–channel complex that senses external stimuli and transmits a local calcium-mediated signal.
- **Diagnosis by US with age-specific criteria:** in first-degree relatives from PKD1 families aged 15–29 years, **≥3 renal cysts confirms** the disease. A cyst-free scan at this age does not reliably exclude it; ultrasonographic exclusion is secure from 40 years (fewer than two cysts). Molecular genetic diagnosis is available.
- **Window for intervention:** defined mutations, dominant inheritance, measurable cyst burden and slow progression permit measures to preserve kidney function.
- **Care is mainly nephrological:** blood pressure control and slowing of renal decline — non-pharmacological (healthy lifestyle, weight management, no smoking) and pharmacological (antihypertensives, statins, analgesics).
- **Tolvaptan (Jynarque):** Food and Drug Administration (FDA)–approved **selective vasopressin V2-receptor antagonist**; slows kidney function decline in adults at risk of rapid progression.
- **Urological referral:** nephrolithiasis, flank pain, gross haematuria, cyst haemorrhage, cyst infection. Haematuria and cyst haemorrhage are usually **self-limiting within days.**
- **RCC risk is not higher than in the general population,** but RCC in ADPKD presents **earlier, is more often sarcomatoid and is harder to diagnose** → haematuria work-up follows guidelines (AUA).
- **Cyst infection:** fever, abdominal pain, raised erythrocyte sedimentation rate or C-reactive protein; **¹⁸F-fluorodeoxyglucose positron emission tomography (PET)** aids confirmation. Treatment is usually conservative with empirical antibiotics — **fluoroquinolones or trimethoprim-sulfamethoxazole.**

- **Intervention is limited, mainly for severe pain from kidney bulk:** procedural (cyst aspiration, embolisation) or surgical (cyst fenestration, nephrectomy). **Routine nephrectomy is not recommended** (morbidity, mortality); selected patients only.
- **ESRD → renal transplantation is the treatment of choice.**

Acquired Cystic Kidney Disease

- **Acquired cystic kidney disease (ACKD):** multiple cysts in ESRD **without hereditary kidney disease**; cysts are thought to arise from the **proximal convoluted tubule, not the distal tubule**.
- **Cystic change tracks the duration of dialysis-dependent ESRD:** up to 80% have cysts by 10 years.
- **World Health Organization (WHO) 2022** classification lists “**acquired cystic disease-associated RCC**” as a distinct entity.

ADPKD VERSUS ACKD

Feature	ADPKD	ACKD
Cause	Inherited: PKD1 (chromosome 16), PKD2 (chromosome 4)	Acquired in ESRD; no hereditary kidney disease
Kidney size	Grossly enlarged , distorted	Small or normal
Relation to ESRD	A cause of ESRD	A consequence of ESRD; rises with dialysis duration
RCC risk	Not increased versus general population	At least tenfold lifetime increase
RCC character	Earlier onset, more often sarcomatoid, harder to diagnose	Multicentric, bilateral, younger, mostly male; mostly indolent; papillary RCC more frequent

Concept — *Bosniak is a wall-and-septum score in which morphology sets the malignancy risk; because cystic RCC is indolent and category III was over-treated, the III–IV decision has shifted towards size-based surveillance. Of the diffuse cystic diseases, **ADPKD causes ESRD and enlarges the kidneys without raising RCC risk; ACKD follows ESRD, leaves the kidneys small and raises the risk at least tenfold.***

KEY POINTS — RENAL CYSTS

- Cysts are the commonest renal mass; most are asymptomatic and incidental.
- The CT-based Bosniak system now includes MRI (version 2019).
- Bosniak I and II need no follow-up; III and IV management includes both surveillance and resection.
- ADPKD: bilateral progressive cystic transformation with slowly progressive CKD.
- ACKD: multiple cysts in small or normal kidneys in ESRD, with increased RCC risk.

ONCOCYTOMA

Epidemiology, Pathophysiology and Natural History

- **The commonest benign enhancing renal mass**, accounting for **up to half of benign SRMs <4 cm**.
- Usually an asymptomatic, unilateral, cortical mass; **hard to distinguish from RCC on imaging, especially chromophobe RCC**.
- **Mostly sporadic**; also occurs in **Birt–Hogg–Dubé syndrome (BHDS)**: autosomal dominant, **folliculin gene on chromosome 17**, variable penetrance → cutaneous **fibrofolliculomas** (face, upper trunk), **pulmonary cysts with spontaneous pneumothorax**, chromophobe RCC and oncocytomas.

- **Renal oncocytosis:** extremely rare, distinct from BHDS; the kidney is diffusely replaced and compressed by numerous bilateral oncocytic nodules → CKD. It may harbour **chromophobe RCC and hybrid oncocytic tumours**, which are not benign and are indistinguishable on cross-sectional imaging.

Evaluation

- **Hypervascular, with a central, sharply defined stellate scar — not specific enough to diagnose oncocytoma or exclude RCC.**
- **Lesion-to-cortex attenuation:** lower attenuation in the corticomedullary phase and more gradual enhancement favour oncocytoma.
- **MRI:** well demarcated; lower T1 and higher T2 signal than cortex. Diffusion-weighted imaging and the apparent diffusion coefficient show promise.
- **Technetium-99m (^{99m}Tc)-sestamibi single-photon emission CT/CT (SPECT/CT):** possible future role; **not yet accurate enough for routine use.**
- **Figure 136.3:** US — 4 cm well-circumscribed hypoechoic mass with a central hyperechoic stellate scar; dynamic CT — heterogeneous enhancement with delayed enhancement of the scar and **segmental enhancement inversion** (highly and less-enhancing segments in the corticomedullary phase reverse in the early excretory phase), a pattern common in oncocytoma.

Management

- **Oncocytoma frequency plus imaging limits** have driven uptake of percutaneous renal mass biopsy.
- **AUA:** biopsy “**should be obtained**” whenever the result may change management (expert opinion). **EAU:** biopsy reduces overtreatment of benign masses and adds grade and subtype for informed decisions (level of evidence 3).
- **But biopsy is unreliable for oncocytoma:** a systematic review found a modest PPV, with a substantial minority proving RCC at surgery; selection bias explains part of this.
- **Two reasons:** inadequate sampling, and **overlapping histology with chromophobe RCC**, which shows intratumoural heterogeneity with oncocytoma-like areas → **EAU: definitive histopathological diagnosis of oncocytoma cannot be made on a needle core.**
- With sufficient representative tissue, **immunohistochemistry with cytokeratin 7 (CK7)** and **cytogenetics** help (table below).
- **Figure 136.4:** well-marginated eosinophilic neoplasm; nested uniform round-to-polygonal cells with dense eosinophilic cytoplasm in hypocellular hyalinised stroma; CK7 single-cell positivity; **KIT positive.**

ONCOCYTOMA VERSUS CHROMOPHOBE RCC

Feature	Oncocytoma	Chromophobe RCC
Behaviour	Benign	Malignant
Imaging	Indistinguishable; stellate scar non-specific; segmental enhancement inversion favours oncocytoma	
Syndrome	Both occur in BHDS (folliculin, chromosome 17)	
CK7	Rarely positive; single-cell pattern	Usually positive, diffuse
Cytogenetics	Limited: normal karyotype, loss of chromosome 1 and/or Y, or 11q13 rearrangement	More extensive abnormalities

Feature	Oncocytoma	Chromophobe RCC
Needle core	Oncocytoma-like areas occur within chromophobe RCC → core cannot give a definitive oncocytoma diagnosis	

Concept — *Oncocytoma is the benign mass that neither imaging nor a core can clear, because its malignant counterpart, **chromophobe RCC**, shares its imaging, histology and syndrome; the diagnosis rests on the CK7 pattern and cytogenetics in adequate tissue.*

KEY POINTS — ONCOCYTOMA

- Benign; up to half of benign SRMs <4 cm; the commonest benign enhancing renal mass.
- May be indistinguishable from malignant masses on imaging.
- Mostly sporadic; also part of BHDS (autosomal dominant, *folliculin*, chromosome 17, variable penetrance).
- Histology overlaps with chromophobe RCC; biopsy misleads unless tissue is sufficient and representative.

ANGIOMYOLIPOMA

Epidemiology, Pathophysiology and Natural History

- **Benign mesenchymal tumour of dysmorphic blood vessels (angio), smooth muscle (myo) and adipose tissue (lipo)**, arising mainly from the renal cortex or medulla.
- **Sporadic or syndromic** — most commonly **tuberous sclerosis complex (TSC)** and **lymphangioleiomyomatosis (LAM)**.
- **Prevalence about 0.13%, female predisposition, peak in the fourth and fifth decades**; mostly asymptomatic and incidental.
- **In TSC, AML occurs in 55–90%**, presenting earlier than sporadic AML.
- Belongs to the family of tumours of **perivascular epithelioid cells**.
- **Driver: biallelic inactivation of TSC2 (tuberin, chromosome 16p13) or, less often, TSC1 (hamartin, chromosome 9q34)**, found in the great majority. Mechanisms: point mutations, small insertions/deletions, large deletions, copy-neutral loss of heterozygosity (commoner in TSC; small mutations commoner in sporadic AML). Multiple tumours in one patient are **clonally independent** (different second hits).
- **TSC is autosomal dominant** with variable penetrance; sporadic mutations are common.
- **Hamartin and tuberin together inhibit the mammalian target of rapamycin (mTOR)**; their loss → unregulated mTOR → cell growth and angiogenesis.

SPORADIC, TSC-RELATED AND EPITHELIOID AML

Feature	Sporadic	TSC-related	Epithelioid
Patient	Fourth–fifth decade; female predisposition	Younger ; no sex predilection	—
Lesions	Usually solitary; smaller	Larger, bilateral, multifocal, recurrent	Minimal fat, abundant epithelioid cells
Symptoms	Mostly asymptomatic	More often symptomatic	—

Feature	Sporadic	TSC-related	Epithelioid
Behaviour	Benign	Benign; retroperitoneal haemorrhage, chronic hypertension, CKD	Aggressive: mitoses, necrosis, venous extension, metastasis, recurrence
Surveillance	US or CT at 6 and 12 months, then annually	MRI every 1–3 years (limits radiation)	—
Treatment	Active treatment if triggers met (Figure 136.6)	Everolimus if >3 cm	Treat as RCC

Evaluation

- **Unlike other benign masses, AML is usually diagnosable by imaging: macroscopic fat on CT or MRI is diagnostic.** AML is described as fat-rich, fat-poor or fat-invisible on imaging.

IMAGING OF AML

Modality	Finding	Pitfall
US	Fat-rich AML hyperechoic	Some RCCs are also hyperechoic → US unreliable
CT	Intralesional fat (–15 to –20 HU) on the non-enhanced series is diagnostic	Fat-poor and fat-invisible AML
MRI	Fat diagnostic; T1 and T2 hyperintensity; gadolinium enhancement; signal loss on fat-suppressed T2; India ink artefact — dark boundary on in-phase/opposed-phase imaging	Same signs in some RCC, lipoma, liposarcoma → weigh enhancement pattern and calcification

- **Very large AML versus retroperitoneal liposarcoma:** AML arises from the renal parenchyma; liposarcoma arises in the retroperitoneum and envelops and compresses the kidney.
- **Percutaneous biopsy** helps in inconclusive or complex cases.
- **Histology:** mature adipocytes without atypia; thick-walled eccentric vessels with spindle cells growing around the lumen; smooth muscle and melanocytic features → **HMB-45 (human melanoma black-45) and Melan-A positive; smooth muscle actin (SMA) positive; keratin negative.**
- **Figure 136.5:** non-contrast CT showing macroscopic fat; T2 hyperintensity; gadolinium-enhanced T1; fat-suppressed T2 with signal dropout.

Management

- **Treatment turns on type (sporadic or TSC), symptoms and spontaneous bleeding risk**, which correlates with size.
- **The historical 4 cm threshold** (in people of childbearing age) was **highly sensitive but poorly specific** for bleeding → many false positives and overtreatment of most patients treated on size alone. A **threshold of >6 cm** has been proposed; bleeding risk rises steeply with size and is highest above 6 cm. **A 4 cm criterion should probably not be a general trigger.** AMLs <4 cm are expected to be asymptomatic and unlikely to bleed → surveillance.
- **Intralesional aneurysm ≥5 mm predicts haemorrhage** → thin-slice CT angiography when observing an AML ≥4 cm.
- **Pregnancy:** bleeding risk may be higher and consequences graver for mother and fetus; **prophylactic intervention is proposed in women planning pregnancy soon** (case reports and small series).
- **Active-treatment prompts:** size, symptoms, residual suspicion of RCC, or epithelioid AML. Four options: embolisation, surgery, thermal ablation, mTOR inhibition.

FIGURE 136.6 MANAGEMENT ALGORITHM FOR RENAL ANGIOMYOLIPOMA (REBUILT)

Situation	Management
Not fat-containing	Differential diagnosis: MRI, biopsy → RCC and other tumours
Spontaneous retroperitoneal haemorrhage	1. Haemorrhagic shock management → 2. Selective arterial embolisation (SAE) → 3. Radical nephrectomy
Symptomatic: intractable pain, haematuria, barriers to health care	Active treatment: 1. SAE; 2. nephron-sparing surgery (NSS) or radical nephrectomy; 3. thermal ablation in selected cases
Asymptomatic/oligosymptomatic, sporadic	Any of: >6 cm, aneurysm >5 mm, growth >0.25 cm/year, childbearing age or pregnant → active treatment. None → active surveillance : US or CT at 6 and 12 months, then annually
Asymptomatic, TSC-related	Everolimus if >3 cm ; MRI every 1–3 years
Asymptomatic, epithelioid	Treat as RCC

From Restrepo et al., 2022 (updating the algorithm of Flum et al., 2016).

- **Trap:** “barriers to health care” is itself an indication for active treatment — surveillance presumes access to imaging and emergency care.

ACTIVE TREATMENT OPTIONS FOR AML

Option	Role	Key points
SAE	First line: symptomatic, prophylactic, pre-surgical, acute haemorrhage	Safe, minimally invasive; high success and emergency bleeding control. Complications: infection/abscess, renal infarction, and post-embolisation syndrome (fever, pain, leucocytosis, nausea/vomiting) — common and self-limiting; major complications rare
NSS (partial nephrectomy)	Preferred surgical approach	Preserves long-term renal function; crucial in TSC (multifocal, bilateral, recurrent lesions); low recurrence; complications as for partial nephrectomy in RCC (open or minimally invasive)
Emergency nephrectomy	Rupture with hypovolaemic shock	When the patient is not an SAE candidate or interventional radiology is unavailable
Thermal ablation	Selected cases	Radiofrequency ablation or cryoablation; small series, acceptable outcomes; limited evidence
mTOR inhibitor (everolimus)	TSC-associated AML; selected sporadic AML	Targets the causative pathway (below)

- **Everolimus** (a rapamycin analogue) is approved by the FDA and European Medicines Agency for TSC-associated renal AML on the basis of the landmark phase III **EXIST-2** trial (TSC- and LAM-associated AML): **AML response (≥50% volume reduction) 42% versus 0% with placebo**; progression uncommon (14.3%); **no patient needed embolisation for bleeding or nephrectomy for growth**.
- **Adverse effects:** stomatitis, nasopharyngitis, acne, proteinuria, headache, cough, hypercholesterolaemia.
- **Cochrane review** (two randomised trials): everolimus much more effective than placebo at ≥50% size reduction (moderate certainty); total adverse events not different.
- **Sporadic AML:** evidence limited to a single-arm cohort of AMLs ≥3 cm — most analysable patients had ≥25% volume reduction, but withdrawal for side effects was frequent.

GUIDELINE POSITIONS: SPORADIC VERSUS TSC-ASSOCIATED AML

Decision	EAU 2023 (sporadic AML)	International TSC Consensus Group 2021
Default	Active surveillance for most, except very large, painful, or ruptured and bleeding tumours	Surveillance of asymptomatic, non-growing lesions
Imaging interval	6–12 months initially , then tailored to growth	MRI every 1–3 years
Acute haemorrhage	SAE (first line)	Embolisation followed by corticosteroids; avoid nephrectomy
Asymptomatic, growing	Size- and risk-based active treatment	>3 cm → mTOR inhibitor first line; SAE or kidney-sparing resection second line
mTOR inhibitor	Weak: selected patients not amenable to SAE or surgery	First line

Concept — AML management is a **bleeding-risk calculation, not a cancer calculation**: size, aneurysm, growth and pregnancy set the trigger and embolisation answers it; in TSC, the loss of mTOR restraint that causes the tumour is also its first-line target, and nephron preservation dominates because the lesions keep coming.

KEY POINTS — ANGIOMYOLIPOMA

- Benign tumour of dysmorphic vessels, smooth muscle and fat; macroscopic fat on imaging is characteristic.
- Mostly asymptomatic, but larger AMLs are a common cause of spontaneous retroperitoneal haemorrhage.
- TSC: multifocal, bilateral AMLs with mTOR activation.
- Treatment individualised by size, symptoms and pregnancy status, aiming to preserve renal function.
- Most managed by surveillance; SAE is the treatment of choice for acute haemorrhage.
- Everolimus for larger, multifocal TSC-related AML and LAM.

PAPILLARY ADENOMA

- **Unencapsulated, low-grade tumour of papillary or tubular architecture, ≤15 mm** (WHO 2016 and 2022), arising from the renal cortex.
- **Nearly always incidental** — on cross-sectional imaging or in nephrectomy specimens removed for larger masses.
- **Incidence rises with age; commoner in men.** Associations with ESRD, acquired cystic disease and sporadic RCC may reflect detection bias.
- **Resembles papillary RCC** microscopically, genetically and immunohistochemically; may be a **precursor**: shared **KMT2C** pathogenic mutation; **trisomy 7 and 17 and loss of Y**, as in papillary RCC.
- **Until 2015, defined as ≤5 mm**; not requiring treatment and often found alongside RCC in nephrectomy specimens.
- **Virtually no metastatic capacity; in a donor kidney, not a contraindication to transplantation.**

Concept — Papillary adenoma is **papillary RCC in miniature** — the same genetics and staining — and its size ceiling, now 15 mm, is what defines it as benign.

KEY POINTS — PAPILLARY ADENOMA

- Low-grade papillary or tubular tumour ≤15 mm; formerly ≤5 mm (until 2015).
- Now mostly incidental on cross-sectional imaging.

- Shares genetics and immunohistochemistry with papillary RCC.
- No metastatic capacity; acceptable in a donor kidney.

METANEPHRIC ADENOMA

- **Metanephric adenoma (MA):** rare benign epithelial tumour, **<0.5% of adult renal neoplasms**; a few hundred reported cases.
- **Early series: nearly half symptomatic** — flank pain, haematuria and **polycythaemia** (erythropoietin and other cytokines). **Recent series: mostly incidental; peak in the fifties; commoner in women.**
- **CT: solid, hypovascular, with prolonged, homogeneous, mild enhancement** — **less than normal parenchyma in all phases and less than typical RCC.** Suggestive but not specific enough to exclude malignancy → often excised, **ideally by NSS.**
- **Biopsy** may help, but its accuracy for MA is not established (a systematic review of renal biopsies contained no MA).
- **Gross:** circumscribed, unencapsulated, solid.
- **Histology mimics papillary RCC:** small epithelial cells in tightly packed small acini, scant cytoplasm, inconspicuous nucleoli, essentially no mitoses.
- **Immunohistochemistry:** **Wilms tumour 1 (WT1) and CD57 positive; CK7 and alpha-methylacyl-CoA racemase (AMACR) negative.**
- **Fluorescence in situ hybridisation (FISH)** for chromosomes 7, 17 and Y: **MA lacks the gains of 7 and 17 and loss of Y** typical of papillary RCC.

Concept — *Metanephric adenoma looks like papillary RCC but does not stain or karyotype like it: WT1 and CD57 positive, CK7 and AMACR negative, without the 7/17/Y changes that papillary adenoma shares with papillary RCC.*

KEY POINTS — METANEPHRIC ADENOMA

- Rare benign epithelial tumour, mostly incidental.
- CT: solid, hypovascular, reduced enhancement — not sufficiently diagnostic.
- Histology resembles papillary RCC; FISH helps: MA lacks +7, +17 and -Y.

MIXED EPITHELIAL AND STROMAL TUMOURS

- **Mixed epithelial and stromal tumour (MEST) and adult cystic nephroma (ACN),** traditionally separate, share epidemiology, imaging and histology → **WHO 2022 places both in the MEST family.**
- The family also contains **paediatric cystic nephroma**, a distinct entity defined by **DICER1** (microRNA-processing gene) mutations — not found in ACN — with downstream effects on p53 function and RAS pathway activation.
- **Middle-aged women, usually incidental; solitary, unilateral, well-circumscribed, fluid-dense, multiseptated masses indistinguishable from cystic RCC** → historically excised, preferably by NSS.
- **MEST histology:** cysts lined by cuboidal and columnar epithelium with occasional urothelium, in a stroma of spindle cells, smooth muscle fascicles and collagen bands. **Stroma: oestrogen and progesterone receptor positive**, plus vimentin, CD34 and desmin; **epithelium CK7 positive.**
- **ACN histology:** variably sized cysts lined by flat, cuboidal and **hobnail** epithelium (nuclei projecting smoothly from the surface); stroma likewise oestrogen and progesterone receptor positive.

- **Malignant transformation has been reported** in both.
- **Manage by current guidelines for cystic renal masses; consider biopsy.**
- **Figure 136.8:** gross — well-circumscribed mass of variably sized cysts separated by thick white septa; microscopy — hobnailed lining and spindle-cell stroma.

Concept — MEST and ACN are **one lesion of middle-aged women with hormone-receptor-positive stroma**; because they image as complex cysts, they are managed through the Bosniak pathway rather than by name.

KEY POINTS — MEST AND ACN

- Uncommon, benign, mostly in middle-aged women; usually incidental.
- One family (with paediatric cystic nephroma) under WHO 2022.
- Solitary multiseptated cystic masses that may resemble cystic RCC.
- Manage per cystic-mass guidelines; consider biopsy.

LEIOMYOMA

- **Very rare benign smooth muscle tumour** of unknown aetiology, mostly incidental; **about 0.3% of primary renal masses; female predominance 2:1.**
- **Commonest origin: the renal capsule**; renal pelvis and renal vein origins also reported.
- **CT:** small, well-circumscribed, homogeneous, exophytic solid mass with uniform enhancement. **MRI:** **hypointense on T1 and T2** (smooth muscle relaxation times), relatively homogeneous enhancement. Still not reliably distinguishable from RCC.
- **Many reported leiomyomas prove to be lipid-poor AML.**
- **Gross:** usually small (mean about 3 cm; up to 7 cm reported); well circumscribed, tan to white, bulging whorled cut surface.
- **Histology:** smooth muscle cells with eosinophilic cytoplasm and **blunt-ended, cigar-shaped nuclei**; no hypercellularity, pleomorphism, mitoses or necrosis. **SMA, desmin and caldesmon positive.**
- **Lipid-poor AML and leiomyoma are mutually misdiagnosed:** both can be HMB-45 positive; **Melan-A is positive in AML but not leiomyoma** — the decisive stain.
- **Management:** per current SRM guidelines (AUA), individualised to symptoms, location, values and preferences. **Biopsy** can supply tissue. If surgery → **NSS preferred** (small, benign, excellent prognosis).
- **Figure 136.9:** small mass arising from the capsule; uniform spindle cells with thin cigar-shaped nuclei; strong SMA staining, absent in adjacent normal tubules.

Concept — Leiomyoma is defined against its mimic: **lipid-poor AML shares its smooth muscle and even HMB-45, and Melan-A decides.**

KEY POINTS — LEIOMYOMA

- Very rare benign smooth muscle tumour, most often from the renal capsule.
- Hard to separate from RCC radiologically; biopsy helps.
- Lipid-poor AML and leiomyoma can be mistaken for each other.
- Manage per solid-mass guidelines; NSS if surgery.

OTHER BENIGN RENAL TUMOURS

- Very rare mesenchymal tumours of unknown aetiology; radiological distinction from RCC is often impossible.

RARE BENIGN RENAL TUMOURS

Tumour	Origin and patient	Presentation and imaging	Distinguishing feature and course
Haemangioma	Vascular neoplasm or malformation; young adults; no sex predilection; any genitourinary organ (testis, cord, bladder, kidney)	Mostly incidental; imaging cannot exclude malignancy → often excised	Sturge-Weber and Klippel-Trénaunay syndromes ; variant anastomosing haemangioma mimics angiosarcoma
Haemangio-blastoma	Indolent mesenchymal tumour, mostly central nervous system (cerebellum)	Radiologically indistinguishable from RCC	Von Hippel-Lindau disease ; sporadic renal cases extremely rare; no recurrence or metastasis after excision
Lymphangioma	Perinephric lymphatics; developmental loss of communication with the lymphatic system	Cystic masses around capsule, sinus or perirenal fat; mostly incidental; obstruction, haematuria, chyluria , hypertension	Aspiration/sclerosis or excision by presentation
Juxta-glomerular cell tumour (reninoma)	Modified smooth muscle cells of the afferent arteriole wall ; women, second-third decade	Refractory hypertension, hyperaldosteronism, hypokalaemia; raised plasma renin ; small (≤ 3 cm) solid mass, may be hypoechoic on US and non-enhancing on CT	Pathognomonic: diffuse granular cytoplasmic renin staining . Excision cures most, not all (hypertensive angiopathy); almost always benign
Renomedullary interstitial cell tumour	Renomedullary interstitial cells	Incidental in nephrectomy specimens or at autopsy; usually <2 cm ; large ones may obstruct	Despite the cells' role in renin release and sodium excretion, no conclusive link to hypertension
Intrarenal schwannoma	Nerve sheath of renal parenchyma; exceedingly rare	Incidental, or flank pain, haematuria, weight loss from slow expansive growth	Benign
Solitary fibrous tumour	Renal mesenchyme; adults, peak fifth decade; no sex predilection	Incidental or mass effect (flank pain, gross haematuria); CT: large, necrosis, calcification, heterogeneous enhancement	Mostly benign; about one in ten aggressive . Malignant predictors: age ≥ 55, size ≥ 15 cm , increased mitoses, necrosis

Concept — Imaging almost never separates the rare mesenchymal tumours from RCC, so each is remembered by one trait: **the syndrome** (haemangioma, haemangioblastoma), **the hormone** (reninoma), **the fluid** (chyluria in lymphangioma) or **the malignant fraction** (solitary fibrous tumour).

KEY POINTS — OTHER BENIGN RENAL TUMOURS

- Rare, unknown aetiology; often indistinguishable from RCC on imaging.
- Haemangiomas may occur in systemic angiomatoses (Sturge-Weber, Klippel-Trénaunay).
- Juxtaglomerular cell tumours secrete renin → hypertension, hyperaldosteronism, hypokalaemia.
- Solitary fibrous tumours can behave malignantly.

DISTINGUISHING MARKERS ACROSS BENIGN RENAL TUMOURS AND THEIR MIMICS

Entity	Positive	Negative / discriminator	Cytogenetics
Oncocytoma	KIT; CK7 rare, single cells	No diffuse CK7	Normal, -1, -Y, 11q13 rearrangement
Chromophobe RCC	CK7 diffuse	—	More extensive
AML	HMB-45, Melan-A, SMA	Keratin	Biallelic TSC2/TSC1 loss
Leiomyoma	SMA, desmin, caldesmon; HMB-45 may be positive	Melan-A negative	—
Papillary adenoma	As papillary RCC	—	+7, +17, -Y; KMT2C
Metanephric adenoma	WT1, CD57	CK7, AMACR	Lacks +7, +17, -Y
MEST / ACN	Stroma: oestrogen and progesterone receptors, vimentin, CD34, desmin; epithelium CK7	—	DICER1 in paediatric cystic nephroma only
Juxtaglomerular cell tumour	Granular cytoplasmic renin	—	—

HEREDITARY SYNDROMES IN THIS CHAPTER

Syndrome	Gene (chromosome)	Renal lesion	Extrarenal
ADPKD	PKD1 (16), PKD2 (4)	Bilateral cysts → ESRD; RCC risk not raised	Liver, pancreas, spleen, ovary, seminal vesicle, arachnoid cysts
Birt-Hogg-Dubé	Folliculin (17); autosomal dominant	Oncocytoma, chromophobe RCC	Fibrofolliculomas; lung cysts, pneumothorax
Tuberous sclerosis complex	TSC1 hamartin (9q34), TSC2 tuberlin (16p13); autosomal dominant	Multifocal bilateral AML	Lymphangioleiomyomatosis
Von Hippel-Lindau	—	Renal haemangioblastoma	Central nervous system haemangioblastoma
Sturge-Weber; Klippel-Trénaunay	—	Haemangioma	Systemic angiomas